

Ankit Patidar

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SUMMARY

Final-year B.Tech CSE (AI & ML) student at VIT Bhopal with strong foundations in machine learning, deep learning, and applied AI. Built and deployed end-to-end ML systems — from satellite image analysis pipelines at INCOIS (Govt. of India) to conversational RAG applications using LangChain, FAISS, and Groq. Comfortable across the ML lifecycle: data processing, model development, evaluation, and API deployment. Actively looking for AI/ML engineering roles where I can contribute meaningfully.

EDUCATION

Vellore Institute of Technology Bhopal

Sep 2023 – Present

B.Tech, CSE (AI & ML) | CGPA: 8.68 / 10

TECHNICAL SKILLS

Languages: Python, C++

ML / DL: Regression, Classification, ANN, CNN, RNN, LSTM, Transformers, Fine-tuning LLMs

GenAI & LLM Stack: LangChain, RAG, Vector Databases, Prompt Engineering, LLM APIs (Groq / OpenAI / Gemini)

API Development: FastAPI, REST APIs, Pydantic, Uvicorn

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV

Tools & Platforms: VS Code, Git, Google Colab, Jupyter Notebook, Docker

EXPERIENCE

Research Intern — INCOIS, Hyderabad

May 2026 – Jun 2026

Ministry of Earth Sciences, Govt. of India

- Built an end-to-end marine macro-litter detection pipeline on EOS-06 OCM (OCEANSAT-3) imagery, processing 13 spectral bands across large-scale coastal zones.
- Engineered 78 spectral indices (FDI, NDVI, PI) with adaptive thresholding and multi-date persistence mapping, reducing atmospheric false positives by ~40%.
- Delivered 5+ production-ready GIS outputs — GeoTIFF layers, NetCDF files, and diagnostic composites — for coastal debris surveillance and prioritisation.

PROJECTS

Marine Macro-Litter Detection — EOS-06 OCM Imagery

Python, Remote Sensing, OpenCV, AI

Research Project — INCOIS, Ministry of Earth Sciences, Govt. of India

- Processed 13 spectral bands of OCEANSAT-3 data, computing 78 normalised difference indices (FDI, NDVI, PI) with adaptive DN-scale thresholding to detect marine plastic debris across coastal zones.
- Applied multi-date persistence mapping across 3+ satellite passes, suppressing ~40% atmospheric false positives; generated GeoTIFF and NetCDF outputs integrated with GIS tools for surveillance reporting.
- Produced 5+ georeferenced deliverables — diagnostic PNGs, composite maps, and GeoTIFF layers — enabling targeted debris collection with significantly reduced manual inspection overhead.

DocuChat — RAG-Powered PDF Chatbot

Python, LangChain, FAISS, FastAPI, Streamlit

Personal Project | github.com/ankitpatidar3739

- Built a conversational RAG pipeline over PDFs using LangChain LCEL, FAISS vector store, and Groq (LLaMA 3.1-8B) with HuggingFace embeddings; page-level source citations on 100% of answers.
- Engineered a 2-stage condense + retrieve + generate pipeline for multi-turn conversation, reducing follow-up query retrieval failure to 0% via standalone question condensation before FAISS lookup.
- Exposed 4 REST endpoints via FastAPI with Pydantic models, file upload validation, and auto-generated Swagger docs; Streamlit UI and FastAPI share all src/ modules with 0 code duplication.

CERTIFICATIONS

- [Applied Machine Learning in Python](#) — Coursera
- [Google Cloud Generative AI](#) — SmartInternz
- [Google IT Support Certificate](#) — Google / Credly

LEADERSHIP & ACHIEVEMENTS

- Management Lead, Music and Band Club — VIT Bhopal